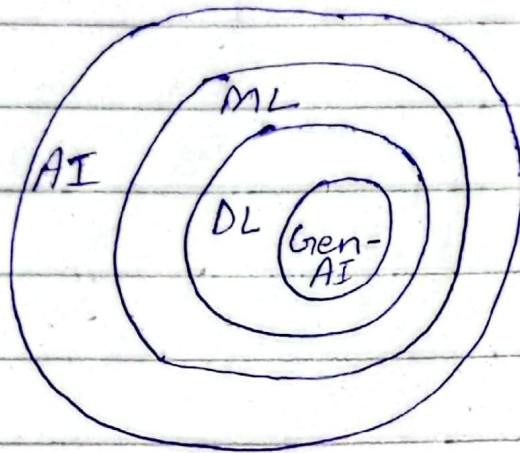


"GEN-AI ROADMAP"

What is GenAI?

Generative AI is a type of AI that creates new content such as text, images, music, or code - by learning patterns from existing data, mimicking human creativity.



Gen AI impacts :

Customer
Support

Content
Creation

Education

Software
Development

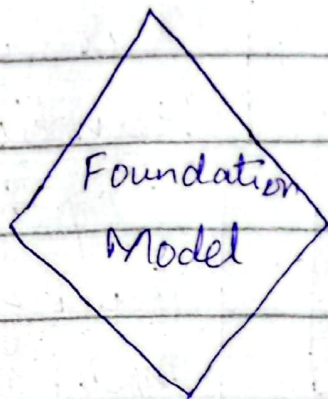
IS GenAI Successful?

- ① It solves real world problems like handling customers, booking tickets online remotely. So yes.
- ② Is it useful on daily basis?
- ③ Is it impacting the world economics?
- ④ Is it creating new jobs?
- ⑤ Is it accessible?

The Problem:

- ① Very Fast progress
- ② The FOMO merchants.
- ③ No single source to learn.

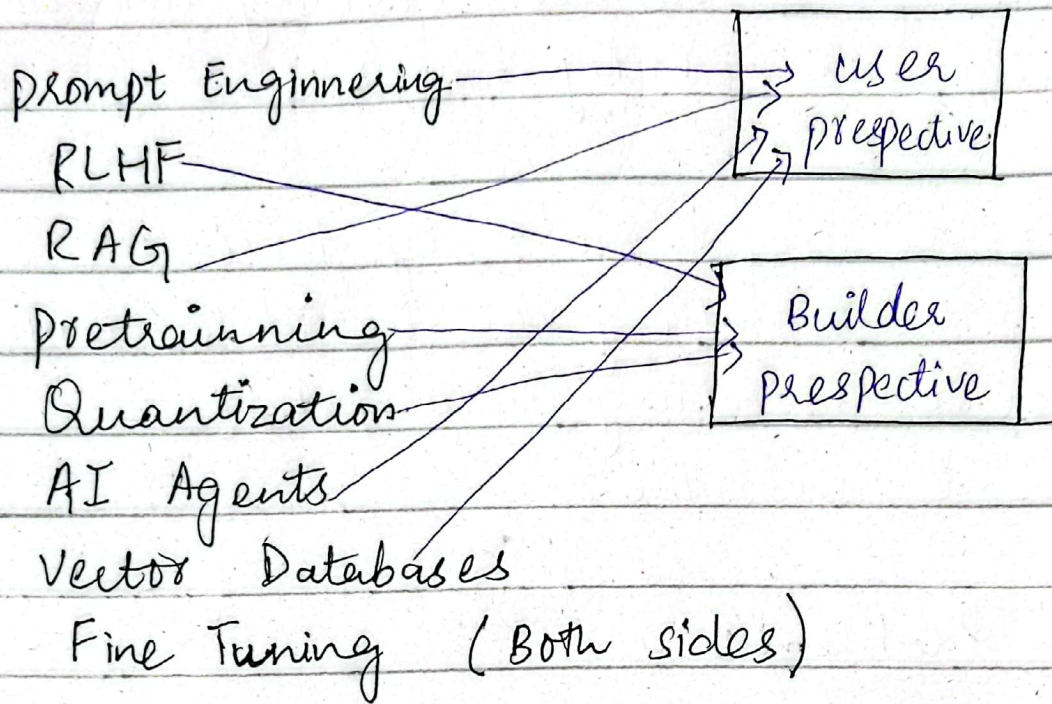
Solution $\Rightarrow$ Mental Model



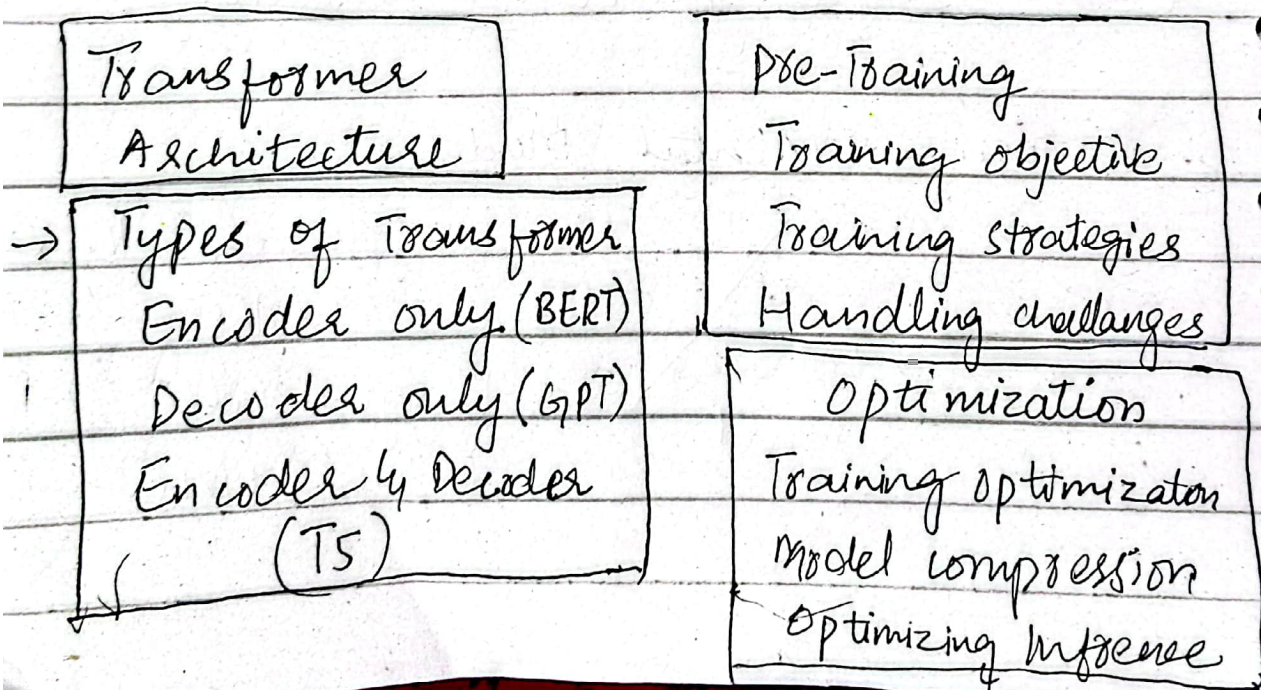
- Can do more than one tasks.
- Huge data
- E.g: LLMs

Two perspective :

- ① use ② Built & deploy.



Builder's Perspective



Fine Tuning	
Task specific tuning	RLHF
Instruction tuning	PEFT
Continual pretraining	

Evaluation	Deployment
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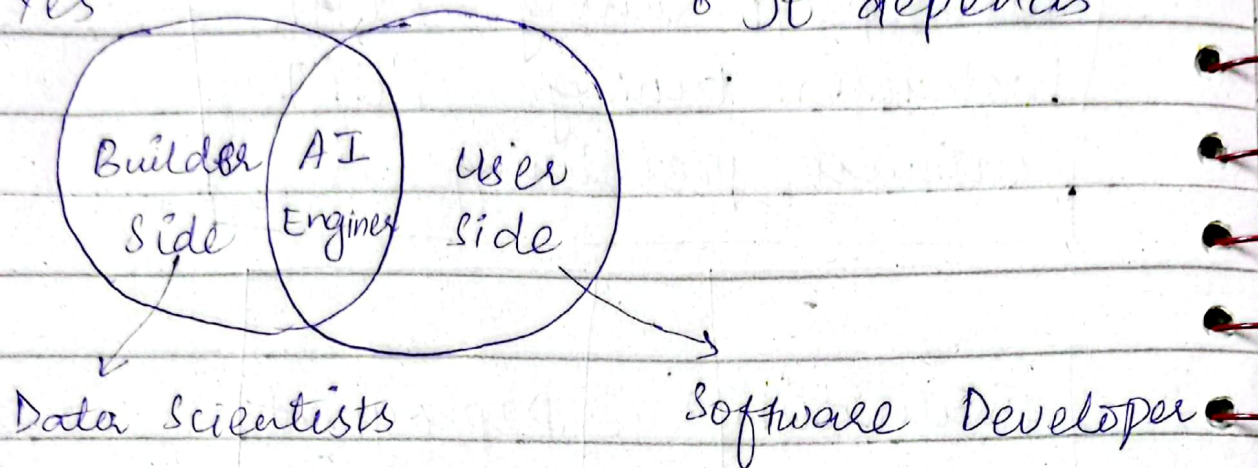
user's perspective

Building Basic LLM Apps	prompt Engineering	Agents
Open Source vs closed	RAG	LLMOps
using LLM API's	Fine Tuning	Miscellaneous
LangChain		
Hugging Face		
Ollama		

Do we need to learn both?

• Yes

• It depends



LEC-2: GenAI using Langchain

What is langchain?

Langchain is an opensource framework that helps in building LLM based applications. It provides modular components and end-to-end tools that help developers build complex A.I applications such as chatbots, ques-ans system, RAG, autonomous agents and more.

- ① Support all major LLMs.
- ② Simplifies developing LLM based applications
- ③ Integrations available for all major tools.
- ④ Open source / Free / Actively developed.
- ⑤ Supports all major GenAI use cases.

Curriculum Structure

